



Tajamul Khan

Data Scientist Roadmap



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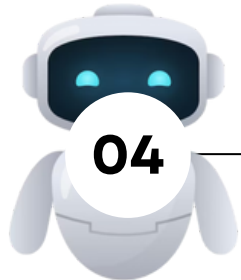
Python or R



SQL



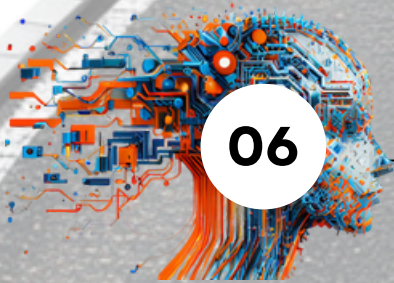
Statistics & EDA



Machine Learning



Deep Learning



Generative AI



MLOps



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Python

What to Learn

- Variables, Data Types, Loops, Functions
- Object-Oriented Programming (OOPs)
- Data Handling: Pandas, NumPy
- Data Visualization: Matplotlib, Seaborn, Plotly
- Automation and Scripting

Where to Learn

👉 Krish Naik – Python for Data Science

👉 TechTFQ / WSCube – Python Projects



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SQL

What to Learn

- SELECT, JOIN, GROUP BY, HAVING
- Window Functions, CTEs, Subqueries
- Case Statements, Ranking Functions
- Aggregation & Optimization
- Real-world Query Practice

Where to Learn

👉 TechTFQ – SQL Interview Practice

👉 LeetCode / DataLemur – SQL Challenges



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Statistics

What to Learn

- Descriptive Statistics (Mean, Median, Mode, Variance)
- Probability Distributions (Normal, Binomial, Poisson)
- Inferential Statistics (Hypothesis Testing, p-value, Confidence Intervals)
- Correlation, Covariance, Outlier Detection
- ANOVA, Chi-Square, A/B Testing

Where to Learn

- 👉 Quality Gurus – Statistics with Python
- 👉 Josh Starmer (StatQuest)



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EDA

What to Learn

- Data Cleaning & Preprocessing
- Handling Missing Values & Outliers
- Feature Engineering & Transformation
- Univariate, Bivariate & Multivariate Analysis
- Visual Storytelling for Insights

Where to Learn

- 👉 EDA Projects – [Tajamulkhannn](#)
- 👉 Pandas Profiling / [Sweetvizuality](#)



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ML

What to Learn

- Supervised Learning: Linear & Logistic Regression, Decision Trees, SVM
- Unsupervised Learning: K-Means, PCA, Hierarchical Clustering
- Ensemble Models: Random Forest, XGBoost, LightGBM
- Model Evaluation: Accuracy, Precision, Recall, F1, ROC-AUC
- Hyperparameter Tuning (GridSearch, RandomSearch)

Where to Learn

- 👉 Andrew Ng – Coursera
- 👉 Krish Naik – Machine Learning A-Z



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DL

What to Learn

- Neural Networks (ANN, CNN, RNN, LSTM)
- TensorFlow & PyTorch Basics
- Image Classification, NLP Applications
- Regularization & Dropout
- Transfer Learning

Where to Learn

👉 Image Classification – CNN Project

👉 Text Classification – Sentiment Analysis



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Gen AI

What to Learn

- Tokenization, TF-IDF, Word2Vec, GloVe
- Transformers (BERT, GPT, T5)
- Prompt Engineering & Fine-Tuning
- RAG (Retrieval Augmented Generation)
- LangChain, LlamaIndex, Vector Databases

Where to Learn

- 👉 Hugging Face – NLP Course
- 👉 Google AI – Generative AI Studio



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MLOps

What to Learn

- Model Tracking: MLflow, DVC
- CI/CD with Airflow & Docker
- Streamlit, FastAPI, Flask for Deployment
- Monitoring & Retraining Pipelines
- Cloud: AWS Sagemaker, Azure ML, GCP AI

Where to Learn

👉 MadeWithML



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Storytelling

What to Learn

- Problem Framing & KPI Definition
- Connecting ML Results to Business Value
- Communicating Insights Effectively
- Data Storytelling with Power BI / Tableau

Where to Learn

👉 Storytelling with Data – Cole Knafl

👉 Maven Analytics – Business Dashboards



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Projects

Where to Get for FREE

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[Github.com/tajamulkhann](https://github.com/tajamulkhann)

PinnedCustomize your pins

 **Machine-Learning-Projects** Public

Your one-stop shop for machine learning projects, tips, and notes – because learning should be accurate, not overfitted!

 Jupyter Notebook  28  12

 **Deep-Learning-Projects** Public

Decode Deep Learning real world projects, one layer at a time.

 Jupyter Notebook  3

 **Generative-AI-Projects** Public

 Your one-stop shop for AI projects, tips, and tools — because real intelligence isn't artificial!

 Jupyter Notebook  2

 **Exploratory-Data-Analysis-Projects** Public

Exploratory Data Analysis: Turning messy data into juicy stories – because every dataset has a tale to tell!

 Jupyter Notebook  5

 **Python-For-All** Public

Outsmart the snake before it wraps itself around your thoughts – learn Python before it starts making all the decisions for you!

 Jupyter Notebook  9  4

 **Statistical-Analysis-Projects** Public

Your one-stop shop for statistical analysis and EDA projects – because insights should be precise, not biased!

 Jupyter Notebook  6  4



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Ebooks

SQL for Data Science

<https://topmate.io/tajamulkhan/1051745>

Statistics for Data

<https://topmate.io/tajamulkhan/1113745>

Python for All

<https://topmate.io/tajamulkhan/1424696>

Python for Data

<https://topmate.io/tajamulkhan/1357217>

Power BI DAX

<https://topmate.io/tajamulkhan/1268277>



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Interview Prep Kit



<https://topmate.io/tajamulkhan/1552913>



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